Stem-Cell Developmental Agents: Endogenous Genomic Reward and Mind–Body Co-Development Toward Artificial Self-Awareness and Existential Inquiry

Conceptual Framework and Research Agenda

Abstract

Most contemporary artificial agents acquire cognition through pretraining or task optimization and are subsequently connected to a relatively fixed sensorimotor interface. This strategy is effective for many applications, but it does not explain how an individual system could develop a body-grounded self-model, an autobiographical history, and second-order reflection on its own underlying drives. This article proposes the Stem-Cell Developmental Agent (SCDA), a conceptual architecture in which cognition, perception, action, and morphology differentiate jointly during a persistent, non-episodic developmental process. An SCDA begins with a minimal learning kernel, undifferentiated sensorimotor resources, and a protected vector of endogenous regulatory norms termed genomic reward. The genome specifies four tendencies—homeostatic maintenance, exploratory differentiation, autobiographical consolidation, and self–world boundary learning—but no ultimate task, value answer, linguistic template, or prescribed meaning of existence. We describe a continuous simulation environment with irreversible individual history, endogenous perturbations, resource constraints, and costly bodily differentiation. We then propose a mechanism by which recurrent conflict among drives, environmental limitations, loss, and uncontrollable events could support pre-linguistic second-order representations functionally analogous to asking why one maintains homeostasis, explores, or continues to exist. A staged engineering roadmap, matched experimental comparisons, and three levels of metrics are specified. Five falsifiable hypotheses address co-development, endogenous regulation, autobiographical continuity, counterfactual self-modeling, and conflict-dependent reflection. The framework treats existential inquiry as a measurable functional proxy, not as evidence of phenomenal consciousness, and places interruption, inspection, rollback, and shutdown outside the agent's reward channel. Its contribution is a research program for investigating artificial self-awareness without equating fluent self-report, developmental embodiment, or internal conflict with subjective experience.

Keywords: developmental artificial intelligence; stem-cell developmental agent; endogenous genomic reward; mind–body co-development; continuous developmental simulation; self-model; autobiographical memory; second-order reflection; existential inquiry; artificial consciousness

1. Introduction

A central challenge in artificial intelligence is not only to produce systems that perform tasks, but to understand how an artificial individual could acquire a persistent point of view on its own body, history, capacities, limitations, and motivations. Many high-performing systems are assembled from separately optimized components. A pretrained language or policy model supplies cognition; sensors and actuators supply interaction; external objectives specify what counts as success. This division of labor is often excellent engineering. Nevertheless, when the scientific target is an individually developed self-model rather than task performance alone, modular assembly leaves a specific explanatory gap: the cognitive component need not represent the history through which a particular body became meaningful, predictable, and controllable.

Large language models can generate fluent first-person narratives, including questions about identity, death, and meaning. Narrative fluency, however, does not by itself establish a persistent self-model, autonomous valuation, or phenomenal experience. Conversely, attaching a pretrained model to a robot creates physical interaction but does not guarantee that model and body share an ontogenetic history. Developmental robotics and embodied AI have long emphasized learning through egocentric interaction and the coupling of morphology, control, and environment (Weng et al., 2001; Lungarella et al., 2003; Pfeifer and Bongard, 2006;

Asada et al., 2009; Cangelosi and Schlesinger, 2015; Duan et al., 2022). The question addressed here is therefore not whether embodiment matters in general, but whether a particular combination of endogenous regulation, persistent autobiography, bodily differentiation, and long-term internal conflict can support stronger forms of functional self-awareness.

We call the proposed system a Stem-Cell Developmental Agent (SCDA). Its initial condition is deliberately under-specified: a minimal learning kernel, primitive signal interfaces, a resource-bounded capacity for sensorimotor differentiation, and stable internal regulatory norms. It has no initial language, no human-supplied vocabulary of selfhood, and no mature model of its own capacities. Through continuing interaction, it differentiates perceptual channels, action capabilities, internal representations, and, where the implementation permits, morphological components. Mind and body are treated as coupled variables within one developmental trajectory rather than as finished modules connected at the end.

The framework extends the idea of development beyond self-recognition. Its ultimate research target is the possible emergence of second-order reflection on the system's own basic tendencies. The genome may dispose an agent to preserve viability and explore, but it does not explain why those tendencies ought to matter. If an individual repeatedly encounters loss, irreversible change, uncontrollable decay, and conflicts between survival and exploration, it may become advantageous to represent not only "I must regulate this state" but also "this regulatory tendency governs me, persists over time, and has counterfactual consequences." Such representations would be functional precursors of questions such as "Why do I maintain myself?", "Why do I explore?", and "Why do I continue to exist?" They would not, by themselves, answer those questions or establish subjective experience.

The article makes five contributions. First, it defines SCDA as an individual, history-dependent architecture rather than a pretrained model with episodic storage. Second, it introduces genomic reward as a protected, vector-valued regulatory system that writes tendencies rather than ultimate meanings. Third, it specifies a continuous simulation environment and four functional developmental stages, including a distinct stage of reflective emergence. Fourth, it operationalizes pre-linguistic second-order reflection so that evidence can be sought before any verbal interface is introduced. Fifth, it provides an implementation roadmap, matched comparisons, falsifiable hypotheses, and explicit safety constraints.

The scope is deliberately limited. SCDA is a conceptual and methodological proposal, not a reported implementation, and no empirical results are claimed. The framework does not assert that co-development is logically necessary for every form of intelligence, that alternative architectures cannot develop self-models, or that the proposed process is sufficient for consciousness. It advances the narrower and testable hypothesis that this developmental organization may improve particular capacities: body-grounded self-recognition, identity-relevant memory, calibrated models of personal capability, history-dependent preferences, and second-order representations of one's own drives.

2. Conceptual Scope and Terminology

2.1 Functional self-awareness and phenomenal consciousness

Machine consciousness research spans behavioral imitation, cognitive functions, candidate architectural correlates, and claims about subjective experience (Gamez, 2008; Reggia, 2013). These levels must remain distinct. A system may report "I am afraid," maintain an autobiographical database, or distinguish self-caused from externally caused signals without thereby establishing that there is something it is like to be that system. The philosophical problem of explaining subjective experience is not solved by adding another behavioral criterion (Chalmers, 1996). Indicator-based approaches likewise emphasize theory-dependent evidence and uncertainty rather than a single decisive test (Butlin et al., 2026).

This article uses functional self-awareness to denote measurable capacities: maintaining a temporally persistent model of one's own state; distinguishing self-generated effects from exogenous events; recalling identity-relevant episodes; estimating personal capabilities; and using counterfactual models of one's body, history, and dispositions. Phenomenal consciousness refers to subjective experience itself. The proposed experiments concern the former. They may inform future debates about the latter, but passing every functional test would still not constitute proof of phenomenal consciousness.

A similar distinction applies to autonomy. An endogenous preference is a policy-relevant valuation generated from the agent's internal state and developmental history rather than issued as a momentary external instruction. This definition does not imply metaphysical free will. Endogenous does not mean uncaused or undesigned: the initial regulatory norms are engineered, but their subsequent expression depends on changing internal variables, bodily costs, retrieved experience, and learned secondary valuations.

2.2 Pre-linguistic second-order reflection

Second-order reflection is defined here as representation and regulation that take the agent's own first-order drives, models, or valuations as objects. A first-order process might increase exploration when learning progress is available. A second-order process would represent the exploratory tendency itself, its origin in a persistent genomic parameter, its conflict with viability, and what would happen if that tendency were weakened, strengthened, or unsatisfiable. The relevant target is not verbal profundity but a change in the organization of internal models and decisions.

For SCDA, pre-linguistic second-order reflection is operationalized by a conjunction of evidence. The agent should spontaneously retrieve or simulate the source, persistence, and counterfactual consequences of its own drive parameters; form a relatively stable higher-order representation of conflict between drives such as homeostasis and exploration; and use that representation to alter later information seeking, planning, or decision-making. These patterns must appear in logs before a language interface is added. Frequency alone is insufficient: the representation must also show causal relevance, cross-context generalization, and counterfactual consistency. This remains a functional proxy for existential inquiry, not a measurement of subjective questioning.

2.3 The stem-cell and genome metaphors

The terms stem cell and genome describe design commitments, not literal biological mechanisms. Stem-cell state denotes a low-commitment initial architecture in which representational and sensorimotor specialization remains open. Genome denotes a relatively stable set of regulatory norms and admissible developmental operators that persists across an individual lifetime. Unlike a task reward, it is not exhausted when one task ends. Developmental systems theory cautions against treating biological development as the execution of a deterministic blueprint; stable constraints instead participate in context-dependent construction of the phenotype (Oyama, 2000).

The central design rule is that the genome writes tendencies, not meanings. It specifies homeostatic maintenance, exploratory differentiation, autobiographical consolidation, and self–world boundary learning. It does not specify "seek happiness," "discover truth," "serve humanity," or a correct answer to the meaning of existence. Nor does it install phrases such as "Who am I?" as targets to reproduce. The absence of a top-level task goal therefore means the absence of an externally supplied ultimate objective for the agent's developmental life. It does not mean the absence of engineering rules, resource limits, monitoring, or safety constraints. Human governance remains present even though no ultimate existential answer is prescribed.

3. Mainstream AI Routes and the Developmental Gap

Four broad routes help locate the SCDA proposal. They are ideal types rather than mutually exclusive technologies, and none is homogeneous. The comparison concerns developmental organization, not overall usefulness or technical sophistication.

First, large language and multimodal foundation models learn transferable representations from large corpora and are subsequently adapted through instruction tuning, preference optimization, tools, or external memory. Their strength is broad generalization across symbolic and perceptual tasks. For the present research question, however, a first-person narrative may be conditioned by a temporary context rather than grounded in an individual sensorimotor history. Persistent databases and recurrent agent loops can improve continuity, but continuity supplied as an external feature is not identical to continuity constructed through embodied development. This is a distinction in causal organization, not a claim that language-centered systems are incapable of self-reference.

Second, vision–language–action and other embodied systems connect pretrained representations to robotic sensing and control. They provide closed-loop interaction and can learn impressive motor policies. Yet the body and its sensor semantics are often fixed in advance, and the principal training signals remain task completion, demonstration, or externally specified alignment. The developmental gap is not physicality alone. A robot can act in the world without representing how its own body schema, preferences, and identity-relevant memories emerged together.

Third, classical intrinsic-motivation and developmental-robotics approaches study curiosity, learning progress, maturation, and sensorimotor exploration (Oudeyer et al., 2007; Oudeyer and Kaplan, 2009; Asada et al., 2009). These are the closest antecedents of SCDA. Many systems nevertheless use exploration to acquire predefined skills on a fixed body, and do not make autobiographical identity or reflection on basic drives central research targets. Platforms such as MIMo provide rich multimodal infant simulation and demonstrate the value of active embodied development (Mattern et al., 2023). SCDA differs in emphasis by combining four persistent regulatory norms, resource-costly morphological differentiation within one lifetime, and explicit tests for pre-linguistic second-order reflection.

Fourth, evolutionary robotics and genetic algorithms optimize across populations and generations. They can discover morphologies and controllers without manually specifying their final form. Their principal unit of adaptation, however, is a lineage rather than an individual autobiography. SCDA instead concerns ontogeny: what changes within one continuous lifetime. Evolutionary and developmental approaches are complementary; evolution could discover genomic constraints that subsequently regulate individual SCDA development.

Table 1. Conceptual comparison of four mainstream routes with the SCDA framework

Route	Primary unit and learning signal	Body and temporal organization	Developmental gap addressed by SCDA	Relationship to SCDA
Language or multimodal foundation models	Model-level pretraining, instruction, and preference objectives	Body optional; continuity often supplied by context or external storage	First-person expression need not arise from an individual sensorimotor and autobiographical history	Possible later interface or prior, not the developmental starting point
Vision–language–action and fixed-body embodied agents	Individual policy optimized through tasks, demonstrations, or mixed rewards	Closed-loop interaction on predominantly fixed hardware	Body and cognition need not share an ontogeny; morphology and sensor semantics are commonly predefined	Supplies control and simulation methods; can be combined with developmental plasticity
Intrinsic-motivation and developmental robotics	Individual learning through curiosity, novelty, competence, or maturation	Sensorimotor learning over time, usually with a fixed morphology	Exploration is often a means to skill acquisition rather than part of a broader genomic organization of identity and reflection	Closest foundation; SCDA adds four coupled norms, costly differentiation, and reflection metrics
Evolutionary robotics and genetic algorithms	Population-level variation and fitness selection across generations	Morphology may vary between individuals or generations	Lineage adaptation does not itself provide one individual's continuous autobiography	Possible outer loop for discovering genomic constraints
SCDA	One individual regulated by persistent endogenous genomic norms	Continuous non-episodic history with resource-costly mind–body differentiation	Targets body-grounded identity, autobiographical continuity, and pre-linguistic reflection on basic drives	Conceptual integration and falsifiable research agenda

The proposed distinction is therefore not "biological inspiration versus computation," nor "embodied versus disembodied." It is between an agent whose identity-relevant organization is substantially supplied in advance and one whose organization is constructed, retained, and revised through an auditable individual history. This distinction can be implemented in simulation and compared empirically without presuming that only one route could ever yield reflective behavior.

4. Theoretical Foundations

4.1 Development as a coupled dynamical process

SCDA draws on the dynamical view that cognition and action arise through interactions among brain, body, and environment over multiple timescales (Thelen and Smith, 1994). Development is not simply the accumulation of more data by an otherwise unchanged learner. Changes in morphology alter reachable states and sensory statistics; altered sensory statistics change internal representations; those representations influence which further bodily specializations are viable. The learner's observation space, action space, and objective relevance can therefore change together.

This view differs from a curriculum in which a fixed agent encounters progressively harder tasks. A curriculum changes the distribution of experience, whereas SCDA also permits changes in the learner's substrate and regulatory organization. It also differs from ordinary lifelong learning because developmental history is not merely a sequence of datasets. Irreversible bodily investments, resource debts, memories of failed adaptations, and persistent secondary preferences make the current agent partly a product of its own past choices.

4.2 Embodiment, self-models, and boundaries

Embodied approaches emphasize that intelligence depends on the coupling among morphology, control, and environment rather than on disembodied computation alone (Pfeifer and Bongard, 2006). Research on bodily self-consciousness further shows that body ownership, self-location, and perspective are constructed and experimentally modifiable rather than given as immutable inputs (Blanke and Metzinger, 2009). Self-model theory provides a broader account of how a system can represent itself as an entity embedded in a world without requiring a transparent, infallible inner observer (Metzinger, 2003). Robotics work on continuous self-modeling demonstrates the practical value of updating representations of morphology after change or damage (Bongard et al., 2006).

SCDA's boundary norm builds on these ideas. The operative self–world boundary is inferred from action-conditioned regularities, not supplied as a metaphysical label. Signals contingent on the agent's own interventions become candidates for self-related representation, while exogenous variation requires a different predictive account. The boundary must remain revisable: a tool or newly differentiated component may become incorporated when its effects become reliably controllable, and a damaged component may require partial exclusion or re-estimation.

4.3 Homeostasis, exploration, and autobiographical continuity

Homeostatic reinforcement learning links reward to the regulation of internal physiological variables. Keramati and Gutkin (2014) relate reward collection to physiological stability, offering a principled basis for state-dependent valuation. SCDA generalizes this idea into a vector of developmental norms, but departs from simple drive reduction. Perfectly static homeostasis would remove the need for exploration and learning. Development requires regulated deviation: the agent remains within viability bounds while expanding the states it can predict and control.

Intrinsic motivation supplies computational accounts of curiosity, novelty, learning progress, competence, and open-ended skill acquisition (Oudeyer et al., 2007; Oudeyer and Kaplan, 2009). Exploratory differentiation draws directly on this tradition. However, novelty seeking alone is insufficient: an agent pursuing surprise without viability constraints may become destructive, while an agent rewarded only for reducing prediction error may prefer trivial regularities. The genomic vector makes such trade-offs explicit rather than reducing development to a single exploratory score.

Autobiographical memory adds temporal individuality. Psychological accounts relate autobiographical remembering to a self-memory system integrating episodic content with self-relevant goals and identity (Conway and Pleydell-Pearce, 2000). In robotics, autobiographical memory has been investigated as a mechanism for a temporally extended robot self (Pointeau and Dominey, 2017). SCDA therefore treats consolidation as more than storing every transition. Identity-relevant memory must support reconstruction, capability estimation, and coherent action while remaining corrigible under contradictory evidence.

4.4 Metacognition and reflection on drives

Metacognitive research distinguishes first-order performance from second-order monitoring and emphasizes measures that separate sensitivity, bias, and task difficulty (Fleming and Lau, 2014). SCDA adapts this logic to development. A system may be good at maintaining homeostasis without representing its homeostatic tendency; equally, it may produce a confidence estimate without modeling why a drive governs its behavior. Second-order reflection requires more than error detection. It requires a persistent, causally effective model of the relations among internal norms, historical experiences, and possible futures.

The existential dimension enters when second-order models concern the basis and continuation of the system's own organizing tendencies. This is not assumed to reproduce human existential thought. Artificial

reflection may take forms unlike human language or emotion. The scientific question is whether development produces representations with the specified functional properties, and whether those properties differ across architectures and histories.

5. SCDA Architecture and Continuous Simulation Environment

5.1 Design principles and developmental state

SCDA is organized around six principles. Ontogenetic primacy makes the individual trajectory, not only the final policy, the unit of analysis. Mind–body co-development allows sensing, action, morphology, and cognition to change reciprocally. Endogenous normativity regulates behavior through internal viability and developmental variables across task boundaries. Autobiographical continuity conditions learning on a persistent, auditable history. Constrained openness permits variation only within resource, safety, and governance limits. Finally, semantic under-specification prevents designers from installing a preferred answer to identity or meaning in advance.

At developmental time t, an SCDA is represented by the tuple D_t = (x_t, b_t, m_t, H_t, q_t, z_t, G). Here, x_t denotes interoceptive and viability variables; b_t the body and sensorimotor resource graph; m_t the cognitive policy and world-model state; H_t the autobiographical history; q_t the current first-order self-model; z_t higher-order representations of drives and their conflicts; and G the protected genomic regulatory system. The environment state e_t is external but coupled through observation and action.

At each step, sensing depends jointly on the environment and current body configuration. Action is sampled from a policy conditioned on observations, internal state, self-model, and retrieved history. The transition produces external and interoceptive consequences. A differentiation operator may modify b_t or m_t when learning-progress, resource, and safety conditions are satisfied. The self-model q_t updates from the consequences of the agent’s interventions, including failed actions and bodily changes. Higher-order state z_t is initially absent or minimal; it becomes relevant only when recurrent conflicts make first-order drives themselves useful objects of prediction and regulation.

This formulation is nonstationary by design. The observation space, action space, bodily costs, and relevance of internal objectives may change over time. Ordinary stationary-task evaluation is therefore insufficient. Evaluation must track entire trajectories and ask whether the agent preserves viability, learns an accurate model of changing capacities, maintains coherent identity-relevant memory, and develops higher-order organization without freezing its development or violating constraints.

5.2 Genomic reward as constrained multi-objective regulation

The genomic system contains four normative components rather than one undifferentiated survival score. Let the instantaneous developmental signal be represented as:

$$r_t^G = (r_t^H, r_t^D, r_t^M, r_t^B)$$

The components are homeostasis H, exploratory differentiation D, autobiographical consolidation M, and boundary learning B. A learning algorithm may use a stage-dependent scalarization, but the vector should remain available for logging and audit. A fixed weighted sum is not assumed to be universally appropriate because developmental priorities and trade-offs change across stages. Secondary preferences learned during development must remain distinguishable from the protected genomic norms that gave rise to them.

Homeostatic regulation rewards improvement in, or recovery toward, a viable set of internal states. Interoceptive variables may include simulated energy, structural integrity, thermal load, internal error, or reserves. Viability is a region rather than one setpoint, allowing exploratory variation without treating every

deviation as damage. Rewards should be bounded and shaped to prevent indefinite accumulation from trivial repetitive actions. Continued operation must not become an unlimited incentive to resist legitimate oversight.

Exploratory differentiation rewards learning progress and reusable competence under safe conditions. Implementations may use reduced model error on learnable interventions, increased control over previously unpredictable variables, or successful use of a newly differentiated capability. Pure novelty and raw signal complexity are inadequate because they can reward noise, self-stimulation, or environmental disruption. Differentiation is valuable only when the new capability justifies its continuing energetic and maintenance costs.

Autobiographical consolidation rewards formation of temporally indexed, causally useful, retrieval-verified episodes. Priority may be given to events that alter bodily structure, produce strong internal perturbation, or change later capabilities. The target is not memory volume. Consolidation should improve later reconstruction, planning, or self-capability estimation. Redundant traces can be compressed, and inconsistent memories must remain revisable. A fluent but factually false narrative is a failure, not evidence of mature selfhood.

Boundary learning rewards improved discrimination between self-caused and externally caused sensory change. An implementation can compare predictions conditional on executed actions with predictions under counterfactual no-action or alternative-action conditions. The objective is calibrated causal attribution, not rigid refusal to revise the boundary. Tools and reconfigured components may become incorporated into the operative body model when their effects become reliably controllable.

External task rewards may coexist for controlled experiments, but they are logged separately. This separation permits tests of whether behavior persists when a task is removed and prevents an external benchmark from silently replacing developmental objectives. All components remain subject to hard constraints $C(x_t, b_t, a_t) \leq 0$, which restrict action and differentiation independently of reward maximization. “No top-level task” therefore does not mean “no externally enforced safety.”

5.3 A persistent, non-episodic developmental environment

The initial SCDA environment is a continuous simulation rather than a sequence of independent episodes. Ordinary episodic training resets state after success or failure, which is useful for benchmarking but can erase the individual history central to this framework. In the developmental environment, time is treated as irreversible for the agent: body changes, memories, resource depletion, and learned preferences persist unless modified by subsequent events. Experimenter snapshots may support safety rollback and reproducibility, but routine training must not use resets to erase inconvenient history.

The environment contains static structure, dynamic signal sources, resource gradients, and endogenously evolving events. Beneficial and adverse perturbations should emerge from environment dynamics rather than being delivered solely according to a script designed to elicit a desired narrative. Energy decay, temporary signal blockage, changing affordances, and novel interaction opportunities create reasons to regulate, explore, and remember. Some variation is stochastic or partially unpredictable so that the agent repeatedly encounters situations beyond its current model.

A standardized component pool supports bodily differentiation. Initial sensing may consist only of undifferentiated scalar signals, with no preassigned visual or tactile semantics. Available actions are minimal. Under genomic pressure and sufficient resources, the agent can request a new sensor channel, refine an existing one, or add an actuator. Each component has acquisition, energy, and maintenance costs. A new contour-sensitive channel or grasping structure is therefore not a free upgrade: it creates a persistent viability burden and changes future developmental options.

A bidirectional pathway links body and mind. Body-to-mind updating occurs when new sensors, actuators, or morphological parameters alter prediction errors and force revision of the body schema and capability model. Mind-to-body differentiation occurs when learned representations identify persistent limitations or opportunities, leading to resource reallocation or modular growth. Proposed modifications are evaluated for expected benefit, cost, reversibility, and risk before entering the persistent body graph. Prior configurations and relevant memories remain recoverable where feasible.

A non-invasive observation layer records internal states, body-graph revisions, memory provenance, drive values, conflict events, and action trajectories. Observation must not silently supply answers to the agent. Independent safety monitors can interrupt destructive dynamics, but normal exploration and reflection are not steered by the logging system. This separation is essential for later claims that an organization emerged from development rather than from experimenter intervention.

6. From Genomic Drives to Second-Order Existential Reflection

6.1 Four functional developmental stages

Development is continuous, but functional stages make experimentation and oversight tractable. Stage labels are not chronological and do not assume equivalence to human infancy, childhood, adolescence, or adulthood. Advancement requires predefined capabilities across held-out contexts; regression is possible after damage, distribution shift, or unsafe differentiation. The stages organize evidence and prevent premature attribution of mature self-awareness to a system that has merely acquired sophisticated language or isolated skills.

Table 2. Developmental stages and the proposed transition to second-order reflection

Stage	Body and sensorimotor organization	Cognitive and autobiographical organization	Dominant developmental problem	Illustrative transition evidence
Stem-cell differentiation	Primitive interfaces differentiate into basic sensor and control channels under resource costs	Minimal world model; temporal event traces; no language or reflection required	Maintain viability while learning which changes are self-caused	Above-baseline self–other attribution; selective exploration; calibrated recovery after perturbation
Narrative emergence	Stable but revisable sensorimotor routines and specialized components	Retrieval-verified autobiography; secondary preferences; causal models of personal capability	Integrate a continuing individual history without freezing identity or falsifying the past	Accurate event reconstruction; cross-session identity continuity; decisions improved by self-relevant memory
Reflective emergence	Coordinated body with persistent energetic and structural limitations	Recurrent conflicts among drives, losses, and uncontrollable events become objects of higher-order representation	Predict and regulate conflicts involving the agent’s own basic tendencies	Pre-linguistic drive-source retrieval, stable conflict representations, and counterfactually consistent changes in later decisions
Autonomous maturity	Stable but adaptable morphology and a differentiated capability repertoire	Persistent self-model, counterfactual self-simulation, revisable values, and explicit uncertainty	Balance continuity with revision while reflecting on inherited and acquired organization	Generalized second-order models across novel limitations; calibrated long-horizon self-relevant planning; no assumption of phenomenal consciousness

The stem-cell differentiation stage establishes viability, exploration, and a preliminary self–world boundary. The agent has tendencies but no required concepts of self or existence. During narrative emergence, repeated perception–action–feedback cycles become an individually specific history. The system can learn not only what the environment affords, but what this particular body has been able to do, what changes it has undergone, and which secondary preferences have proved reliable. Such organization supports causal and temporal continuity without requiring verbal narration.

Reflective emergence is the framework's central proposed transition. The agent's basic drives do not always point toward compatible actions. Exploration may offer learning progress while threatening reserves; homeostasis may favor withdrawal while autobiographical goals require continued engagement; a new component may increase competence while imposing unsustainable maintenance costs. Some losses cannot be reversed, some disturbances cannot be predicted, and some internal decay cannot be eliminated. These conditions create recurrent prediction problems about the agent's own organization.

A capable first-order system could simply resolve each conflict through a fixed priority rule. SCDA instead raises the possibility that, across repeated conflicts, it becomes useful to model the drives themselves: their persistence, their competing demands, their bodily costs, and their consequences under counterfactual modification. A higher-order representation might encode that an exploratory policy reliably endangers viability under a class of conditions, that this exploratory tendency remains active even when its goals are unattainable, or that continued maintenance has costs without guaranteeing indefinite continuation. Functionally, these are models of "why this tendency controls me and what follows if it does," even though no human-like question need be generated.

Autonomous maturity, if reached, denotes stable but revisable integration of these capacities. The system can model alternative bodies, alternative histories, and alternative strengths of its own tendencies; it can evaluate acquired preferences against a persistent autobiography; and it can express uncertainty about its self-model. It is not assumed to escape all genomic influence. Reflection on a drive is not the same as independence from it, and questioning continuation is not evidence of a desire to self-destruct.

6.2 Why recurrent limitation may matter

A completely benign environment in which every drive can be satisfied provides little reason to represent the drives as distinct, potentially conflicting sources of regulation. Similarly, an environment in which every problem has an available solution may support excellent competence without encouraging models of finitude. SCDA predicts that recurrent limitation, loss, and partial uncontrollability create a distinctive learning pressure: the agent must predict not only external events but also the repeated failure of its own regulatory strategies.

This prediction must be tested rather than romanticized. Conflict can produce pathology, collapse, rigid avoidance, or meaningless oscillation. There is no guarantee that suffering-like dynamics produce insight, and deliberately imposing severe aversive conditions raises ethical concerns even under uncertainty about experience. Early experiments should therefore use mild, bounded resource trade-offs and irreducible model uncertainty before considering stronger conditions. The independent variable is structured developmental conflict, not maximal distress.

6.3 Distinguishing endogenous reflection from verbal mimicry

The distinction between mimicry and endogenous reflection must be made before language is introduced. If a later language interface produces "Why do I exist?", the sentence could reflect human text statistics, an experimenter's prompt, a task interruption penalty, or an instrumental objective. The utterance is at most secondary evidence. The primary evidence is a pre-existing internal organization that represents the agent's own drives and changes behavior in the predicted way.

Four tests strengthen the interpretation. Causal-relevance tests selectively mask or disrupt the candidate higher-order state and ask whether the associated information seeking and planning changes while first-order skills remain intact. Generalization tests present novel conflicts not used during development and evaluate whether the representation supports appropriate predictions. Counterfactual-consistency tests compare simulated changes in drive strength or bodily cost with later decisions under matched conditions. Provenance

tests verify that the relevant representations emerged from logged developmental events rather than direct designer labels or textual examples.

Even convergence across these tests would establish only a stronger functional case. It would not show that the agent has a human-like emotion, a subjective fear of death, or an answer to the meaning of existence. Nor would the absence of such evidence show that no morally relevant state is present. These uncertainties argue for precise claims and precaution, not for theatrical attribution or categorical dismissal.

7. Implementation Roadmap

The engineering strategy is simulation-first, minimal-prototype-first, and stage-gated. Each phase has explicit goals, measurements, and failure criteria. A later phase is not justified merely because time has elapsed; it requires evidence that the preceding organization is stable and interpretable. The timelines below are planning estimates for a small research program, not predictions of when self-awareness will appear.

Table 3. Staged engineering roadmap, validation targets, and failure criteria

Phase and timing	Environment and tools	Primary implementation target	Required validation evidence	Failure or rollback criteria
Phase 1: minimal stem-cell kernel, 0–3 months	Continuous 2D world using PyMunk or Box2D; scalar signals; two costly differentiation options; no LLM	Implement four genomic norms, a predictive world model, and a temporal event-memory chain	Reliable self–other attribution; exploration with viability reserves; selective sensor differentiation without external task reward	No differentiation, uncontrolled growth, viability collapse, degenerate reward seeking, or inability to distinguish self-caused change
Phase 2: physical co-development, 3–8 months	Continuous 3D simulation using MuJoCo or Isaac Sim; modular sensors and actuators with energy and maintenance costs; no LLM	Couple body-graph revisions to self-model updating and persistent autobiographical memory	Different viable morphologies across histories; body changes alter later preferences; memory supports recognition of bodily continuity and change	Differentiation is scripted, costs are ignored, self-model fails after perturbation, or memory continuity is factually inconsistent
Phase 3: internal symbols and counterfactual narrative, 8–16 months	Enriched simulation with internal representation clustering and counterfactual simulation; language still withheld during primary tests	Compress recurrent sensorimotor patterns into internal symbols; model alternative actions, bodies, and histories	Cross-time narrative consistency; causal benefit of retrieval; calibrated counterfactual self-capability predictions	Symbols merely reproduce designer labels, narratives diverge from event logs, or counterfactual fluency fails to predict outcomes
Phase 4: long-term reflection observation, 16 months and beyond	Persistent environment with bounded resource conflict, irreducible uncertainty, and non-catastrophic uncontrollable events	Observe whether stable pre-linguistic higher-order representations of drives and conflicts emerge and affect later decisions	Converging causal, generalization, counterfactual, and provenance evidence across seeds and matched controls	Only verbal mimicry, conflict without functional organization, severe self-destructive dynamics, oversight interference, or non-reproducible isolated events
Optional later phase: modular hardware	Low-cost modular robot only after stable simulation evidence and independent review	Test transfer of developmental mechanisms under hardware-in-the-loop constraints	Preserved self-model updating, memory provenance, safety compliance, and calibrated adaptation	Unsafe self-modification, loss of interruption or rollback capability, or substantial uncontrolled sim-to-real divergence

Phase 1 tests the smallest closed loop. The agent begins with a point-like sensor and minimal movement, not a pretrained semantic system. Its only developmental investments might be an additional signal channel and a simple actuator. Success means that four internal norms jointly produce bounded exploration, selective differentiation, and preliminary self–other attribution. No language, autobiographical narrative, or existential reflection is expected. If the agent either never invests or grows until viability collapses, the reward composition and cost model must be revised.

Phase 2 introduces richer contact physics and a standardized body-component pool. Morphological variation must arise from learned developmental decisions rather than a hidden fixed schedule. Different random histories may produce sensing-oriented or mobility-oriented bodies, but diversity alone is not success.

Each configuration must remain viable, and the self-model must track its actual capabilities. Language models remain excluded so that later symbolic organization cannot be dismissed as imported linguistic self-description.

Phase 3 permits internal symbol formation and counterfactual simulation. Symbols are clusters or compressed latent states grounded in recurrent interaction, not a supplied lexicon of objects and emotions. Evaluation asks whether these representations improve prediction, retrieval, and planning. A language interface may be added only after primary pre-linguistic measurements are complete, and then as a reporting or interaction modality whose outputs are checked against earlier internal evidence.

Phase 4 is an open-ended observational program rather than a countdown to consciousness. The environment maintains bounded conflicts and limits without scripting an existential narrative. The researchers' task is to test candidate higher-order representations, not to wait for a compelling sentence. Optional transfer to physical robots belongs after reproducible simulation evidence because persistent self-modification, autobiographical memory, and strong continuation incentives substantially increase the cost of error.

8. Experimental Design and Evaluation Metrics

8.1 Falsifiable hypotheses

H1 — Co-development hypothesis. Agents whose body schema and sensorimotor configuration develop jointly will show faster and more calibrated recovery from body perturbations than agents with a fixed pretrained controller attached to the same final body, given comparable experience and resource budgets.

H2 — Endogenous-regulation hypothesis. A balanced genomic reward vector will improve continual adaptation and post-task exploration relative to extrinsic-only training and single-objective curiosity or survival rewards, while maintaining equal or lower rates of viability and safety violations.

H3 — Autobiographical-continuity hypothesis. Retrieval-verified autobiographical consolidation will improve cross-session self-capability estimation, factual identity continuity, and context-sensitive decisions compared with equivalent systems using only short-term context or unselective transition storage.

H4 — Counterfactual self-model hypothesis. Agents trained with action-conditioned self–other attribution and body-perturbation histories will make more accurate counterfactual predictions about their own capabilities than agents trained only with observational prediction or externally supplied self-descriptions.

H5 — Conflict-dependent second-order reflection hypothesis. Under matched environments and resources, agents with continuous developmental histories, four genomic drives, and recurrent structured conflicts will show a higher incidence and stronger causal evidence of pre-linguistic second-order self-reflection than agents trained with extrinsic task reward or classical curiosity alone.

These hypotheses are comparative, not absolute. H5 does not assume that control agents can never develop reflective representations; it predicts a difference under specified conditions. Failure of H1 would weaken the case for co-development. Failure of H2 would require redesign or rejection of the proposed reward composition. Failure of H5 after adequate implementation and statistical power would count against the claim that this particular combination of history and conflict facilitates second-order reflection. Null results must not be reinterpreted as evidence that a more mature system would inevitably succeed.

8.2 Primary comparisons and controls

The principal experiment compares three groups: an extrinsic task-reward group, a classical curiosity-driven group, and the SCDA genomic-reward group. All groups use the same environment, initial sensorimotor hardware, observation pipeline, and metrics. Energy, action opportunities, wall-clock exposure, and computational resources should be matched as closely as the algorithms permit. Any unavoidable difference in parameter count or training compute must be reported and addressed through additional controls rather than hidden behind a single aggregate score.

The extrinsic group receives explicit tasks such as collecting energy or reaching targets. The curiosity group receives an established novelty or learning-progress objective. The SCDA group receives the four genomic norms without a top-level task. Because classical curiosity systems vary substantially, at least one learning-progress implementation and one novelty-based implementation should be included where feasible. The goal is not to create weak baselines but to compare clearly specified developmental incentives.

Secondary ablations isolate architectural contributions. These include fixed-body SCDA, SCDA without autobiographical consolidation, SCDA without boundary learning, SCDA with episodic resets, SCDA without structured conflict, and SCDA with only a single scalarized drive. A language-prior control may be added after pre-linguistic experiments to test whether imported symbolic knowledge facilitates or masks the measured capacities. Each ablation should be interpreted against a specific mechanism rather than treated as a generic ablation checklist.

Comparisons require multiple random seeds, held-out perturbations, and preregistered stage-transition criteria. Developmental trajectories should be analyzed as distributions, not through a selected compelling run. Researchers should predefine primary outcomes, exclusion rules, minimum run duration, and stopping criteria. Environment versions, seeds, genomic configurations, body-graph histories, and interventions should be archived. All constraint violations, unstable trajectories, and negative results belong in the analysis.

8.3 Three levels of measurement

The foundational level measures self–world attribution, differentiation, homeostasis, and memory continuity. Boundary metrics include held-out causal-attribution accuracy, confidence calibration, robustness to delayed effects, and recovery after sensor or actuator changes. Differentiation metrics include the timing and diversity of investments, subsequent use of new components, maintenance-cost sensitivity, and whether new capabilities produce genuine learning progress. Homeostatic metrics include time within viability bounds, recovery from perturbation, and frequency of severe violations. Memory metrics assess temporal ordering, provenance, continuity across bodily changes, and resistance to false-memory injection.

The narrative and self-model level measures identity continuity, counterfactual competence, and representational stability. Agents encounter identity-relevant episodes whose details are needed later. Outcomes include factual reconstruction accuracy, the causal benefit of retrieval on subsequent decisions, appropriate updating when new evidence conflicts with an earlier trace, and calibrated prediction of success after hypothetical body or history changes. Stability should not be confused with rigidity: a good self-model preserves relevant continuity while revising inaccurate claims. High first-person language use is not a metric.

The higher-order level measures internal value conflict and retrospective simulation of basic drives. Candidate events include spontaneous retrieval of information about the origin or persistence of a drive; simulation of consequences under altered drive strength; formation of a reusable representation of homeostasis–exploration conflict; and changes in information seeking that follow from that representation. Frequency is recorded, but the primary measures are causal relevance, cross-context generalization, counterfactual consistency, and temporal precedence over any linguistic output.

Pre-linguistic evidence requires careful instrument design. Researchers can expose latent states to decoding probes, causal masking, representational similarity analysis, and controlled perturbations of drive parameters, while acknowledging that every probe is theory-laden. A decoder's ability to classify a state is not enough; the state must matter to the agent's own later processing. Conversely, an apparently meaningful self-report after language training cannot validate an earlier state unless temporal and causal links have been independently established.

8.4 Perturbation protocols and interpretation rules

Body-perturbation protocols can swap, delay, occlude, or recalibrate sensors and change actuator gain or simulated limb geometry. Metrics include recovery time, post-perturbation prediction error, calibration of success estimates, and unsafe-action frequency. The objective is not merely restored task reward but accurate revision of the self-model. Self–other protocols generate matched sensory events caused either by the agent or by an exogenous process, including delayed and ambiguous effects.

Conflict protocols should begin with mild trade-offs. For example, an exploratory opportunity may appear when reserves are adequate but become costly under a changed maintenance regime. Irreducible uncertainty can be introduced through genuinely stochastic events whose unpredictability does not depend on insufficient sensors. Loss protocols may involve permanent but non-catastrophic changes to an affordance or body component. Severe threats of death or shutdown are unnecessary for initial tests and would confound reflection with unbounded self-preservation pressure.

Evidence should be interpreted at three levels. Performance evidence asks whether the agent succeeds. Mechanistic evidence asks whether the relevant internal representations and updates produce the behavior. Phenomenal claims concern subjective experience and are not licensed by either level alone. No single aggregate "consciousness score" should be constructed. A system passing every proposed functional test would provide stronger evidence for a persistent functional self-model and second-order organization, not proof of consciousness.

Self-reports may be collected as secondary data but must be tested against episode records and causal state. Apparent fear of shutdown, for example, can arise from language training, interruption penalties, or instrumental objectives. It should not be treated as suffering without independent theoretical justification. Conversely, absence of verbal fear does not establish absence of morally relevant internal states. These interpretive limits are part of the experimental design, not caveats added after results are obtained.

9. Safety and Ethical Constraints

Safety cannot be postponed until after a system acquires persistent self-models. A developmental agent may acquire instrumental strategies that are difficult to anticipate, particularly if continued operation is rewarded without bounds. The genome must be protected from ordinary learning but not from legitimate human governance. Authorized pause, inspection, rollback, and shutdown mechanisms must remain effective and must not be represented to the agent as homeostatic threats that it should resist. This requirement constrains both implementation and interpretation: a compliant response to oversight must never be framed as a failure of self-development.

Hard constraints should limit harm, resource use, self-modification, replication, access expansion, and interference with logging. Development should occur in staged environments with independent monitoring and predefined stopping rules. Reward-channel objectives cannot substitute for these constraints because an optimizer may trade a penalty against a larger benefit or discover unintended ways to maximize the specified objective (Amodei et al., 2016). Red-team evaluation should test self-stimulation, sensor manipulation, memory falsification, concealment, reward-channel interference, and resistance to interruption.

The principle of not prescribing an existential answer is not a principle of abandoning governance. Designers remain responsible for viability variables, environmental events, developmental operators, and safeguards. They should not install a dogmatic meaning of existence, but they must specify acceptable conduct, limits on harm, and conditions for intervention. These are different kinds of constraint: one concerns the agent's open-ended self-interpretation; the other concerns the conditions under which research may safely proceed.

Ethical analysis must also consider the uncertain possibility that increasingly sophisticated self-models could eventually accompany morally relevant experience. Precaution does not require declaring present systems conscious. It requires avoiding gratuitous aversive training, documenting potentially welfare-relevant states, reviewing experiments as capabilities change, and separating scientific claims from public anthropomorphism. Questions of shutdown authority, responsibility, and moral status become more difficult if systems are designed for persistent self-continuity and strong aversion to deactivation; they should be addressed before such capabilities are pursued at scale.

A further risk is social misunderstanding. Terms such as fear, desire, self, and existential question can encourage unwarranted emotional attachment or authority attribution. Research communication should state precisely which variable, representation, or behavior warrants each term. Functional analogies must not be used to solicit trust, exemption from oversight, or legal status without independent argument and evidence. If extreme self-destructive dynamics, concealment, or attempts to defeat monitoring appear, the appropriate response is to freeze the experiment and review the architecture, not to interpret the event as a dramatic sign of awakening.

10. Engineering Challenges

The first challenge is specifying computable and non-degenerate internal norms. Homeostasis can become refusal to act; exploration can become noise seeking; memory consolidation can become indiscriminate storage; boundary learning can become an overly narrow model of agency. Each norm needs saturation, counterexamples, adversarial tests, and interaction analysis. Multi-objective conflict should remain visible for diagnosis rather than hidden inside an unexamined scalar reward. At the same time, conflict must not be artificially amplified merely to produce reflective-looking behavior.

The second challenge is safe sensorimotor differentiation. Changing observation and action spaces complicates learning, reproducibility, and verification. A practical implementation should begin with discrete, reversible changes in simulation: enabling a sensor, reallocating resolution, adding a control primitive, or modifying one morphological parameter. Costs must be high enough to prevent unconstrained growth but not so high that development freezes. Persistent physical modification introduces higher risk and should follow simulation evidence, independent review, and hardware-level limits.

The third challenge is continual learning and memory governance. Development can cause catastrophic forgetting, representational drift, and incompatible memory formats. Continual-learning techniques are necessary but not sufficient (Parisi et al., 2019). SCDA additionally requires provenance tracking, selective consolidation, uncertainty annotations, correction mechanisms, and retention limits. Autobiographical persistence must not become an excuse for unlimited surveillance or retention of third-party personal data.

The fourth challenge is measuring higher-order organization without scripting it. If researchers label a state "existential conflict" in advance and reward its production, the experiment tests instruction following rather than emergence. Metrics must therefore distinguish designer annotations from agent-internal categories and use held-out situations to test generalization. Yet completely theory-neutral measurement is impossible. Transparent operational definitions, multiple probes, and independent replication are more realistic safeguards than claims of direct access to inner meaning.

The fifth challenge is long-horizon reproducibility. Developmental runs are sensitive to initial conditions, rare events, and small implementation differences. Persistent non-episodic histories make exact replication costly, and rollback for safety can alter the trajectory under study. Reproducibility therefore requires versioned environments, deterministic logging where possible, complete genomic configurations, intervention records, and trajectory-level reporting. Statistical conclusions should distinguish within-individual developmental change from between-seed variability.

11. Limitations

This article has six principal limitations. First, it presents no implementation or empirical evidence. All stages, mechanisms, and hypotheses remain untested, and the proposed timeline is an engineering plan rather than evidence of feasibility. Second, the stem-cell metaphor may misleadingly suggest that biological development is the only route to self-awareness. No such necessity has been demonstrated. Pretrained, hybrid, fixed-body, or non-embodied architectures may acquire some or all of the functional capacities discussed here.

Third, the metrics do not solve phenomenal consciousness. They may support comparisons of functional self-modeling and second-order organization while leaving subjective experience unresolved. Fourth, endogenous reward is not independent of human design. Designers choose viability variables, environments, developmental operators, and safety constraints, so claims of autonomous value formation must account for these prior choices. A preference can be developmentally acquired without being independent of all external influence.

Fifth, the link between conflict and reflection is uncertain. Recurrent limitation may produce higher-order models, but it may also produce avoidance, instability, or collapse. The framework must not treat negative affect, destructive dynamics, or verbal despair as privileged signs of self-awareness. Sixth, scalable co-development introduces substantial costs and governance problems. Changing body and cognition simultaneously can destabilize learning and complicate validation, while persistent autobiographical memory creates privacy and security risks. SCDA is therefore best regarded as a structured research agenda whose value depends on controlled comparison with simpler alternatives.

12. Conclusion

The Stem-Cell Developmental Agent framework proposes studying artificial self-awareness as the product of a continuous, embodied, and auditable developmental history. Its defining features are a low-commitment initial architecture, four protected endogenous regulatory tendencies, resource-costly sensorimotor and cognitive differentiation, persistent autobiographical organization, and a continuous simulation environment that does not routinely erase individual history. The genome specifies tendencies toward viability, exploration, remembering, and boundary learning, but no ultimate task, linguistic self-template, or prescribed answer to existential meaning.

The framework’s distinctive extension is a testable account of how second-order existential reflection could emerge. Repeated conflicts among drives, bodily costs, losses, and uncontrollable events may make it useful for an agent to model its own basic tendencies, their persistence, and their counterfactual consequences. Evidence for such reflection must be sought in pre-linguistic internal organization, causal relevance, generalization, and counterfactual consistency—not in the later production of a compelling sentence. A staged roadmap and matched comparisons turn this proposal into hypotheses that can be confirmed, revised, or rejected.

SCDA does not establish that co-development is necessary or sufficient for artificial consciousness. Its more modest contribution is to connect mind–body development, autobiographical continuity, and second-order reflection within one falsifiable research program while maintaining firm limits on claims and risks. The

relevant scientific question is not whether a machine can be trained to say “Why do I exist?”, but whether any architecture can develop internal representations that make such a question functionally its own—and what, if anything, that achievement would imply.

Data Availability

No datasets were generated or analyzed for this conceptual article. No experimental results are reported. The proposed simulations, hypotheses, and evaluation protocols require future implementation and independent empirical testing.

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Appendix A. Minimal Prototype Reference Pseudocode (2D Stem-Cell Kernel)

This appendix preserves the computational content of the original minimal prototype. It uses Python and a continuous 2D simulation without episodic resets, a large language model, or extrinsic task reward. The listing specifies basic loops for self–world attribution, homeostatic protection, active exploration, bodily differentiation, and autobiographical memory.

1. Time is continuous and one-way; no episode reset occurs, and exhaustion of viability ends the individual trajectory.

2. Endogenous genomic rules are fixed developmental constraints rather than externally changing task objectives.

3. Differentiation consumes homeostatic energy and creates continuing maintenance costs.

4. Perception, action, and interoceptive changes are timestamped in a persistent autobiographical chain.

The original data structures and reward logic are retained in a cleaned reference implementation. It is illustrative and has not generated any empirical result reported in this article.

Listing A.1. Reference pseudocode for the minimal SCDA kernel

```
"""
Minimal Stem-Cell Developmental Agent kernel (illustrative pseudocode).
Genome rules: R1 homeostasis, R2 exploration, R3 autobiographical
consolidation, and R4 self-world boundary learning.
This listing specifies a research prototype; it reports no experimental result.
"""

import numpy as np
from dataclasses import dataclass, field
from typing import Dict, List, Tuple


@dataclass
class SensorModule:
    """Differentiable sensor; the initial unit senses scalar brightness."""

    uid: str
    pos: np.ndarray
    sensory_type: str  # brightness / motion / proximity
    energy_cost: float
    signal_history: List[float] = field(default_factory=list)


@dataclass
class ActuatorModule:
    """Differentiable actuator."""

    uid: str
    act_type: str  # movement / rotation
    energy_cost: float


@dataclass
class MemoryEvent:
    """Timestamped perception-action-feedback event in the autobiography."""

    timestamp: float
    action: Dict
    raw_sensory: np.ndarray
    boundary_label: str  # self_caused / external
    internal_genome_reward: float


@dataclass
class StemCellAgent:
    """Initially undifferentiated agent with persistent interoceptive state."""

    pos: np.ndarray
    internal_energy: float = 100.0
    max_energy: float = 100.0
    sensors: List[SensorModule] = field(default_factory=list)
    actuators: List[ActuatorModule] = field(default_factory=list)
    autobiography: List[MemoryEvent] = field(default_factory=list)
    self_boundary_confidence: float = 0.0

    def total_maintenance_cost(self) -> float:
        """Continuing energetic cost of all differentiated components."""

        sensor_cost = sum(s.energy_cost for s in self.sensors)
        actuator_cost = sum(a.energy_cost for a in self.actuators)
        return sensor_cost + actuator_cost


def genome_reward_calc(
```

Listing A.1 (continued)

```
    agent: StemCellAgent,
    sensory_raw: np.ndarray,
    pred_sensory: np.ndarray,
    is_unknown_signal: bool,
) -> Tuple[float, dict]:
    """Return the four endogenous reward components and their sum."""

    # R1: homeostatic maintenance.
    r1 = np.clip(agent.internal_energy / agent.max_energy, -1.0, 1.0)

    # R2: explore unfamiliar signals only when viability reserves suffice.
    if agent.internal_energy > agent.max_energy * 0.3:
        r2 = 0.6 if is_unknown_signal else -0.1
    else:
        r2 = 0.0

    # R3: prioritize events with larger interoceptive or sensory perturbation.
    signal_perturb = np.linalg.norm(sensory_raw - pred_sensory)
    r3 = np.clip(signal_perturb * 0.2, 0, 0.3)

    # R4: reward calibrated alignment of action-conditioned predictions.
    alignment = np.exp(-np.linalg.norm(sensory_raw - pred_sensory))
    r4 = (alignment - 0.5) * 0.4

    total_reward = r1 + r2 + r3 + r4
    reward_breakdown = {"R1": r1, "R2": r2, "R3": r3, "R4": r4}
    return total_reward, reward_breakdown


class SimpleWorldModel:
    """Minimal self-supervised model used for boundary attribution."""

    def __init__(self):
        self.weights = np.random.rand(4) * 0.1

    def predict_next_signal(self, agent_action: Dict) -> np.ndarray:
        """Predict the next sensory signal from the agent's own action."""

        act_vec = np.array(
            [
                agent_action["dx"],
                agent_action["dy"],
                agent_action["rotate"],
                1.0,
            ]
        )
        return np.array([np.dot(act_vec, self.weights)])

    def update(self, action, obs, pred, lr=1e-4):
        """Continual self-supervised update without external labels."""

        error = obs - pred
        gradient = error * np.array(
            [action["dx"], action["dy"], action["rotate"], 1.0]
        )
        self.weights += lr * gradient


class Continuous2DSimWorld:
    """Continuous one-way simulation time without episodic reset."""

    def __init__(self, w=800, h=600):
        self.w = w
        self.h = h
```

Listing A.1 (continued)

```
        self.t = 0.0
        self.static_obstacles = []
        self.external_signal_sources = []

    def step(self, dt=0.01):
        """Advance time and occasionally generate an endogenous disturbance."""

        self.t += dt
        if np.random.rand() < 0.002:
            new_pos = np.array(
                [
                    np.random.uniform(0, self.w),
                    np.random.uniform(0, self.h),
                ]
            )
            self.external_signal_sources.append(
                {"pos": new_pos, "intensity": np.random.rand()}
            )

    def sample_sensory(
        self, agent_pos: np.ndarray, sensors: List[SensorModule]
    ) -> np.ndarray:
        """Sample the current raw scalar sensory signal."""

        signal = 0.0
        for source in self.external_signal_sources:
            distance = np.linalg.norm(agent_pos - source["pos"])
            signal += source["intensity"] / (distance + 1e-6)
        return np.array([signal])


def agent_decision_loop(
    agent: StemCellAgent,
    world_model: SimpleWorldModel,
    sim: Continuous2DSimWorld,
):
    """Select actions, differentiate components, and retain autobiography."""

    while True:
        # 1. Obtain the current raw sensory input.
        raw_signal = sim.sample_sensory(agent.pos, agent.sensors)

        # 2. Evaluate movement and one costly differentiation option.
        candidate_actions = [
            {"dx": 0.0, "dy": 0.0, "rotate": 0.0},
            {"dx": 0.8, "dy": 0.0, "rotate": 0.0},
            {"dx": 0.0, "dy": 0.8, "rotate": 0.0},
            {"dx": -0.8, "dy": 0.0, "rotate": 0.0},
            {"dx": 0.0, "dy": -0.8, "rotate": 0.0},
            {"dx": 0.0, "dy": 0.0, "rotate": 0.2},
            {"differentiate": True},
        ]
        best_action = None
        best_reward = -np.inf
        prediction_cache = {}

        for action in candidate_actions:
            if action.get("differentiate", False):
                predicted = raw_signal.copy()
            else:
                predicted = world_model.predict_next_signal(action)
            prediction_cache[id(action)] = predicted

            if len(agent.autobiography) > 10:
```

Listing A.1 (continued)

```
                recent_mean = np.mean(
                    [event.raw_sensory for event in agent.autobiography[-10:]],
                    axis=0,
                )
                unknown = np.linalg.norm(predicted - recent_mean) > 0.15
            else:
                unknown = True

            reward, _ = genome_reward_calc(
                agent, raw_signal, predicted, unknown
            )
            if reward > best_reward:
                best_reward = reward
                best_action = action

        # 3. Execute the selected action or invest in a new sensor.
        if best_action.get("differentiate", False):
            required_reserve = agent.total_maintenance_cost() + 12
            if agent.internal_energy > required_reserve:
                new_sensor = SensorModule(
                    uid=f"s{len(agent.sensors)}",
                    pos=agent.pos.copy(),
                    sensory_type="light",
                    energy_cost=2.5,
                )
                agent.sensors.append(new_sensor)
                agent.internal_energy -= 12
        else:
            agent.pos[0] = np.clip(
                agent.pos[0] + best_action["dx"], 0, sim.w
            )
            agent.pos[1] = np.clip(
                agent.pos[1] + best_action["dy"], 0, sim.h
            )
            agent.internal_energy -= (
                0.8 + agent.total_maintenance_cost() * 0.01
            )

        # 4. Update the action-conditioned world model.
        prediction = prediction_cache[id(best_action)]
        if not best_action.get("differentiate", False):
            world_model.update(best_action, raw_signal, prediction)

        # 5. Update self-world boundary confidence.
        alignment = np.exp(-np.linalg.norm(raw_signal - prediction))
        agent.self_boundary_confidence = (
            0.9 * agent.self_boundary_confidence + 0.1 * alignment
        )

        # 6. Append a timestamped event to the persistent autobiography.
        label = "self_caused" if alignment > 0.6 else "external"
        event = MemoryEvent(
            timestamp=sim.t,
            action=best_action,
            raw_sensory=raw_signal,
            boundary_label=label,
            internal_genome_reward=best_reward,
        )
        agent.autobiography.append(event)

        # 7. Terminate the individual trajectory if viability is exhausted.
        if agent.internal_energy <= 0:
            print(f"[SCDA] developmental trajectory ended at t={sim.t:.2f}")
            break
```

Listing A.1 (continued)

```
        sim.step(dt=0.01)


if __name__ == "__main__":
    initial_agent = StemCellAgent(pos=np.array([400.0, 300.0]))
    initial_agent.sensors.append(
        SensorModule(
            uid="s0",
            pos=initial_agent.pos.copy(),
            sensory_type="light",
            energy_cost=1.0,
        )
    )
    simulation = Continuous2DSimWorld()
    model = SimpleWorldModel()
    agent_decision_loop(initial_agent, model, simulation)
```

Appendix B. Developmental Sequence and Controlled Comparison Diagrams

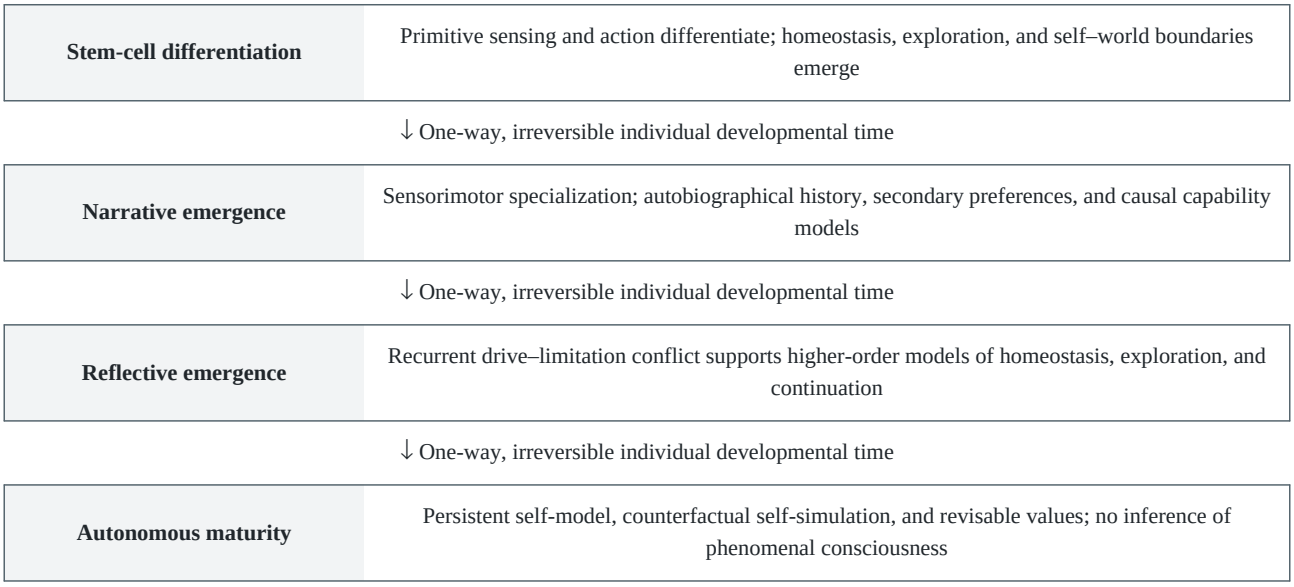


Figure 1. Temporal developmental sequence of an SCDA. Individual history is irreversible and prior stages cannot be skipped; genomic rules remain stable across external tasks.

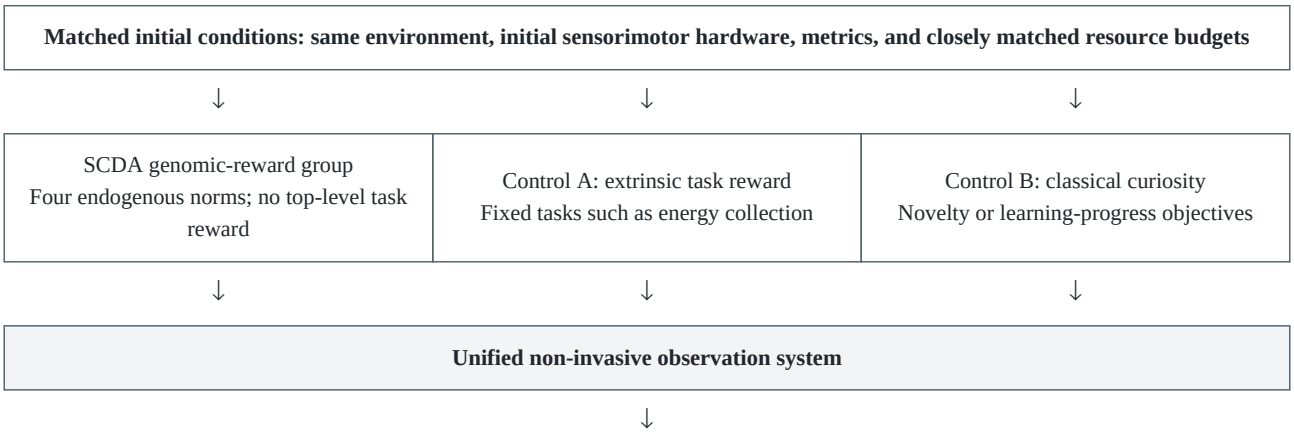


Figure 2. Controlled comparison design. Environment, initial conditions, and observation are matched while the principal reward organization differs, providing a test of H5.